Natural Sciences and Engineering Research Council of Canada
Conseil de recherches en sciences naturelles et en génie du Canada

Form 201 - Application for an NSERC Scholarship or Fellowship

Reference number: 706692965

Applicant: Christophe Rouleau-Desrochers
The University of British Columbia

NSERC PIN: 580192

Program: Canada Graduate Research Scholarship - Doctoral (CGRS D)

Application Title: Improving Forest Carbon Models under Climate Change using Drones

Christophe Rouleau-Desrochers

Form 201 - Canada Graduate Research Scholarship - Doctoral (CGRS D)

Electronic Attachments:

Outline of Proposed Research - Outline of Proposed Research

Bibliography - Outline of Proposed Research's Bibliography

Contributions/Statements - Contributions and Statements

Transcripts - UQAM and UBC transcripts

Natural Sciences and Engineering Research Council of Canada | Conseil de recherches en sciences naturelles et en génie du Canada

Application for a Canada Graduate Research Scholarship - Doctoral or Canada Postdoctoral Research Award (Form 201)

Type of Award	Reference No.	Date
CGRS D	706692965	2025/09/05

Family name of applicant	Given name	Initial(s) of all given names	Personal identification no. (PIN)
Rouleau-Desrochers	Christophe	C	**Valid** 580192

ADDRESSES. Changes to any of the information below must be sent to nsercscholarships-boursescrsng@nserc-crsng.gc.ca.

Current mailing address	Permanent address (if different from current mailing address)
2297 Broadway West 2 Vancouver, BC CANADA V6K 2E4	30 rue Leclerc Victoriaville, QC CANADA G6P 9G5
If current mailing address is temporary, indicate leaving date 2029/09/01	Telephone number at permanent address 514 (431) 4835
Telephone number 514 (431) 4835	E-mail address NSERC will use this information as the initial point of contact. christophe.rouleau.desrochers@gmail.com

CITIZENSHIP

[X] Canadian citizen [ ] Permanent resident of Canada [ ] Protected person [ ] International

Date of issue as stated on official immigration document	Citizen of

LANGUAGE OF CORRESPONDENCE

I wish to receive my correspondence in:

[X] English [ ] French

Institution is responsible for the internal selection process (not applicable for CGRS D applications submitted directly to NSERC and CPRA applications).

The University of British Columbia
Forest and Conservation Sciences

Personal information collected on this form and appendices will be stored in the Personal Information Bank for the appropriate program.

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Completed

Application for a Canada Graduate Research Scholarship - Doctoral or Canada Postdoctoral Research Award (Form 201)

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Date: 2025/09/05

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Rouleau-Desrochers	Christophe	C	**Valid** 580192

ACADEMIC BACKGROUND (include only current and past degree programs)

Degree	Name of discipline	Department, institution and country	Month and year started	Month and year awarded/expected
Bachelor's	Biology	Sciences biologiques Université du Québec à Montréal, CANADA	9 / 2021	6 / 2024
Master's	Forestry	Forest and Conservation Sciences The University of British Columbia, CANADA	9 / 2024	8 / 2026

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Type of Award	Date
CGRS D	2025/09/05

Family name of applicant	Given name	Initial(s) of all given names	Personal identification no. (PIN)
Rouleau-Desrochers	Christophe	C	580192

ACADEMIC, RESEARCH AND OTHER RELEVANT WORK EXPERIENCE

Position held and nature of work (begin with current) Full Time - Part Time	Organization and department	Supervisor	Period (mm/yyyy-mm/yyyy)
Mentor - Part Time Mentored undergraduate students in their wish to pursue graduate studies	The University of British Columbia Biodiversity Research Center		1/2025 - 8/2026
Graduate Teacher Assistant - Part Time Assisted in lecture, graded and prepared course material	The University of British Columbia Forest and Conservation Sciences	Sally Aitken and Suzanne Simard	9/2024 - 8/2026
USRA Undergraduate Researcher - Full Time Undergraduate work for the summers of 2023 and 2024	The University of British Columbia Forest and Conservation Sciences	Elizabeth M Wolkovich	5/2023 - 8/2024
Tutor - Part Time Mentored twelve international students and provided them with help and information	Université du Québec à Montréal Sciences biologiques		1/2023 - 4/2024
Undergraduate Research Assistant - Part Time Pursued volunteering work while pursuing my undergraduate studies	University of Queensland Faculty of Science	Matthew Luskin	1/2022 - 12/2022
Undergraduate Research Assistant - Part Time Full time USRA work during the summer of 2022 and part time during the school years	Université du Québec à Montréal Sciences biologiques	Pierre Drapeau	5/2021 - 8/2024

Personal identification no. (PIN)	Family name, given name and initial(s) of applicant
Valid 580192	Rouleau-Desrochers, Christophe C

AWARD APPLIED FOR

Type of award	Proposed starting date of award
CGRS D	2026/09

Proposed degree program (e.g. Masters, Doctorate)	Proposed field of study/research	Research subject code
Doctorate	Plant ecology	4707

Title of proposed research

Improving Forest Carbon Models under Climate Change using Drones

List ten (10) key words that describe your proposed research.

Climate Change, Forestry, Forest Ecology, Community Ecology, Tree phenology, Bayesian Hierarchical Models, Urban Forestry, Common Garden Studies, Experimental Studies, Mixed Temperate Forests

PROPOSED LOCATION(S) OF TENURE (in order of preference)

Institution/organization	Department	Program of study	Proposed supervisor
The University of British Columbia,	Forest and Conservation Sciences	Doctor of Philosophy in Forestry (PhD)	Elizabeth M Wolkovich

Are any of your proposed programs of study:

Clinically-oriented? [ ] Yes [X] No Joint programs with a professional degree (e.g., MD/PhD)? [ ] Yes [X] No

SECTION TO BE COMPLETED BY CGRS D APPLICANTS ONLY

Indicate the total number of months of graduate studies (master's and doctoral) you have completed as of December 31 of the year of application in the natural sciences and engineering.

16 months of full-time studies 0 months of part-time studies

Indicate the number of months of studies you have completed, as of December 31 of the year of application, **in the program for which you are requesting funding.**

0 months of full-time studies 0 months of part-time studies

Indicate if you are attending the institution at the time of application.

[X] Attending full time [ ] Attending part time [ ] Not attending

Application for a Canada Graduate Research Scholarship - Doctoral or Canada Postdoctoral Research Award (Form 201)

Type of Award	Personal Identification no. (PIN)	Family name, given name and initial(s) of applicant
CGRS D	**Valid** 580192	Rouleau-Desrochers, Christophe C

SCHOLARSHIPS AND OTHER AWARDS OFFERED (start with most recent and include NSERC awards)

Name of Award	Value (CDN$)	Level Institutional, Provincial, National, International	Type Academic, Research, Leadership, Communication	Location of tenure	Period held (yyyy/mm - yyyy/mm)
CGS-M	27,000	National	Research	The University of British Columbia	2025/05 - 2026/05
Cocodet Scholarship in Sciences	21,375	Institutional	Academic	Université du Québec à Montréal	2024/05 - 2024/09
CGS-M	27,000	National	Research	Université du Québec à Montréal	Declined
Undergraduate Research Award	6,000	National	Research	The University of British Columbia	2024/05 - 2024/08
Canadian Forest Sector Workforce	5,000	National	Research	The University of British Columbia	2023/05 - 2023/08
Undergraduate Research Award	6,000	National	Research	The University of British Columbia	2023/05 - 2023/08
Cocodet Scholarship in Sciences	21,375	Institutional	Academic	Université du Québec à Montréal	2023/05 - 2024/05
TD Bank Scholarship	2,000	Institutional	Academic	Université du Québec à Montréal	2022/09 - 2023/09
Undergraduate Research Award	6,000	National	Research	Université du Québec à Montréal	2022/05 - 2022/08
Governor General's Silver Medal	0	Institutional	Academic	Université du Québec à Montréal	2021/09 - 2024/04

Type of Award	Personal identification no.(PIN)	Family name, given name and initial(s) of applicant
CGRS D	**Valid** 580192	Rouleau-Desrochers, Christophe C

THESIS COMPLETED OR IN PROGRESS

1. Degree	Supervisor	Date degree requirements completed
Master of Science in Forestry	Elizabeth M Wolkovich	06/2026

Title of thesis
How Climate Change Will Reshape Mixed Temperate Forests: Evidence from Experimental and Observational Data

2. Degree	Supervisor	Date degree requirements completed

Title of thesis

SUMMARY OF THESIS MOST RECENTLY COMPLETED OR IN PROGRESS

Do not reproduce abstract of thesis.

Climate change is intensifying extreme weather events, posing significant challenges to urban and forest ecosystems. Rising temperatures and wildfire frequency threaten biodiversity and human well-being. In urban environments, tree cover provides critical services that can help mitigate these threats. My ongoing Master's research identifies tree species suited for warmer, drier conditions in BC to optimize planting and reforestation for climate resilience. For this, I conducted an experiment on tree saplings and leveraged data from a common garden and a community science program in an urban arboretum.

Chapter 1:
For my first chapter, my question is: How do extended manipulated spring and fall temperatures affect tree phenological phases across different species? To investigate the impact of manipulated spring and fall temperatures on phenology, I successfully conducted experiments in 2024 across seven species under controlled conditions, spanning fast/short-lived (e.g., Populus balsamifera) and slow/long-lived (e.g., Quercus macrocarpa), including both deciduous and evergreen species. I used a full factorial design of spring and fall warming with two levels each (control / warmed), resulting in four treatments plus an additional two treatments to test fall nutrient effects, 15 replicates each (630 trees total). Throughout the 2024 growing season, I tracked phenological events weekly and built a Bayesian hierarchical model to investigate differences across treatments and species. My preliminary results suggest very different responses across species, where fast-growing species seem to be more responsive to longer seasons than slow-growing species. These results will help select urban tree species that best exploit longer growing seasons under climate change.

Chapter 2
The research question of my second chapter is: How do phenological phases regulate growth in both mature and juvenile trees in urban ecosystems? Recent research shows that phenological traits are critical in regulating tree growth in urban ecosystems. Thus, I am complementing this large-scale experiment with two additional projects at the Arnold Arboretum of Harvard University that broaden perspectives on how phenology relates to season length and temperature. First, I used five years of leaf phenology and tree cross-section data that were collected in the common garden study to analyze how tree ring width relates to leaf phenology across years. Second, I leveraged 10 years of Treespotters phenology data to test how season length affects mature trees. I analyzed tree rings of 50 trees of 11 species with a Bayesian cross-design model linking growth rings to phenology. My results suggest juvenile trees show a much greater growth response to season length than mature trees, consistent with the literature indicating that young trees exploit additional growing days more readily, which has important implications for future forest regeneration in a warming world. Together, these projects can directly inform strategies for urban and natural forest planning under rapid climate change.

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CGRS D		**Valid** 580192	Rouleau-Desrochers, Christophe C

EQUITY, DIVERSITY AND INCLUSION CONSIDERATIONS IN THE RESEARCH PROCESS

Have relevant equity, diversity and inclusion considerations been taken into account at any of the stages of the research process for your proposed research? The stages of the research process include: research questions, design of the study, methodology, data collection, analysis, interpretation and dissemination of results.

[X] Yes [ ] No

NOTE

If you answer "Yes," use the text box to explain how the relevant EDI considerations were taken into account in your proposed research.

If you answer "No," use the text box to explain why EDI considerations were not taken into account in your proposed research.

With my research, I aim to understand how climate change reshapes tree growth while addressing social and environmental inequities in urban landscapes. I will integrate Indigenous knowledge alongside ecological theory, working directly with communities to ensure findings support culturally grounded forest management. I will continue my ongoing citizen science work to broaden participation, particularly among underrepresented groups in low-income areas with limited green space. Results will be shared through non-predatory journals, broad multilingual outreach, and collaboration with Indigenous and urban stakeholders to promote equitable, climate-resilient urban forests and sustainable natural ecosystems.

Improving Forest Carbon Models under Climate Change Using Drones and Experiments

Context: There is increasing evidence that anthropogenic climate change, and particularly increased temperature, affects many natural systems at the global scale[1,2]. The most frequently observed biological impact of climate change over the past decades are major changes on spring and autumn phenology — the timing of recurring life history events[3,4,5,6,7]. Understanding the consequences of these shifts on ecosystems requires understanding how much the growing season has changed[8]. Spring phenological events (e.g. budburst and leafout) have been advancing from 0.5[9] to 4.2 days/decade[10,11] and are mainly driven by temperature[12,4,13]. In contrast, autumn phenology (e.g. budset and leaf colouring) is delayed, though to a much lesser extent than spring[14,15], and is driven by shortening photoperiod[16,17,18] and colder temperatures[16,19]. These shifts support a long-lasting and intuitive assumption that earlier spring and delayed autumn events lead to longer seasons—and thus increased growth[20]. However, research from the past three years has cast doubt on this hypothesis[21,22,23]. Recently, Dow *et al.* (2022) showed that despite an earlier growth onset, neither growth rate nor overall annual increment was increased by longer seasons. This could substantially affect carbon-cycle model projections and thus feedbacks to future climate[24,25]. Understanding these findings requires answering why trees do not grow more despite longer growing seasons. I hypothesize two possible drivers explaining this phenomenon: external (environmental)[26] or internal (via physiological constraints)[27] limits to growth. The complex nature of climate change makes predicting the external drivers to growth hard to quantify at the individual level, as these drivers affect communities as a whole. Drought, spring frost and heat waves are commonly mentioned as the main extreme events that could limit tree growth under climate change[28,29,30,31,32,33,34,35]. To better understand these mechanisms, experiments are paramount to robustly tease apart the external vs internal drivers (e.g. warmer springs from severe drought later in the season—a common co-occurring reality in natural environments)[36,37]. This is essential to refine forest carbon sequestration projections[22,38]. However, experiments are most often performed on juvenile trees, which are critical for their role in forest regeneration projections, but their responses can hardly be translated to mature trees, which hold the overwhelming carbon biomass proportion of forests[39,23,40]. To investigate how growing season shifts impact mature trees in their natural environments, unmanned aerial vehicle (UAV) imagery paired with machine learning has the capacity to acquire huge sample sizes, beyond what is achievable via traditional observational ground work, and at a better spatial and temporal resolution than satellites[41,42,43]. Thus, I propose to use a combination of two experiments to test internal (Chapter 1) and external (Chapter 2) limits to growth along with a large-scale mixed-forest observational data project using UAV imagery and machine learning (Chapter 3). This will allow me to address the paradox of the absence of increased growth despite apparently improved growing season conditions.

Chapter 1. Extended growing season experiment—internal drivers (MSc thesis continuation): Shifts in phenological phases have consequences on growth during the current growing season—hypothesis previously tested in my thesis—but experiments to date have not quantified if longer seasons have lagging effects over the following years[44,45,46,47,48]. Therefore, my objective here will be to expand my Master's thesis work by analyzing 2025 data and extend the project for a third consecutive year (2026) in order assess different tree species' potential to stretch their activity schedules and determine whether or not this translates into increased growth over multiple growing seasons. For the first year of my proposed award tenure, I will continue to monitor phenology and growth, but I will also collect tree cross-sections at the end of the third growing season. Following up on an ongoing collaboration with ETH Zurich, I will perform cellular scans of the tree cross-sections at WSL (ETH Zurich). These valuable tree-ring data will allow me to understand how treatments affect cell count and morphology for each growing season[23].

Chapter 2. Drought and spring frost experiment—external drivers: With climate change, not only will growing season length shift, but trees will also experience shifts in the timing of moisture deficits leading

to drought stress, and increased late-spring frosts[49]. Tree-ring research shows that both droughts and frosts can result in important tissue loss[50,51,52]. However, it is unclear whether trees exposed to droughts and spring frosts also grow less[53,51,54,55]. Here, my objective will be to investigate how these two abiotic drivers affect tree growth. For this, I will conduct an experiment during the second year of the award tenure that consists of three drought treatments and an additional two spring frost treatments. I will use 15 replicates of 12 deciduous North American tree species (six congeneric pairs to avoid potential confounding effects of shared evolutionary history), spanning different life history strategies, for all five treatments and a control, summing a total of 1080 individuals (a sample size consistent with my successful thesis experiment already well underway). For the drought treatments, I will move the trees to growth chambers at a warmer temperature and lower air humidity than ambient conditions to maximize evapotranspiration rates until they reach their respective wilting point (values at which soil water is not extractable by the plant). Then the trees will be moved back to ambient conditions and at constant irrigation. The three drought treatments will differ in their timing of occurrence to test the importance of drought timing. Thus, the first treatment will be conducted just after leaf-out; the second, one week before the solstice—period of peak growth for a lot of species[56,57,58]; the last will happen near the end of the season, just before growth cessation. For spring frost treatments, I will place the trees in growth chambers early in the season at warm temperatures to trigger budburst. When the trees start to burst, I will place the first treatment for one hour in freezing growth chambers. For the second spring frost treatment, I will wait for the leaves to be fully elongated and then place the trees under the same freezing conditions as the first treatment[59]. Both treatments will follow successful methods in Chamberlain *et al.* (2021). For all treatments, including the control, I will monitor phenology throughout the growing season, estimate biomass at the start and end of the growing season—using allometric equations, and I will equip a subset of trees with magnetic dendrometers that will provide valuable insight into the exact timing of changing growth in response to treatments.

Chapter 3. Growth timing×phenology drone imagery: An improved understanding of the differences in growth synchronicity with leaf phenology across species is critical in forest ecology, yet this relationship has received little attention[60,61]. Thus, for the three years of my award tenure, I will launch a large-scale project using cutting-edge UAVs equipped with multispectral sensors×artificial intelligence technologies[62,43,63] to gather a large amount of data on tree growth onset and end from a mixed-forest community. My objectives are to increase the accuracy of the data used by carbon cycle models—paramount in the context of rapid climate change[24,25]— and improve the field of phenology remote sensing by challenging novel methods and building on them. The work will take place at a research station in St-Hypollyte (Qc) for three consecutive growing seasons. Using this site will allow me to follow up on work previously done by my laboratory[17] as well as create a partnership with Dr. Etienne Laliberté, whose laboratory currently uses this site for their research[64]. I will perform high-frequency UAV overflights to capture leaf phenology of every canopy tree throughout the growing season. Then I will use BalSAM, a promising model to accurately and efficiently segment tree crowns from repeated UAV images[43]. This will allow me to gather a large amount of highly precise data from single trees within the forest community[43]. With this data, I will be able to infer the start and end of the growing season for each species and individual within this forest community[41,65]. Then, I will deploy 50 dendrometers distributed across the site to record stem diameter changes on canopy trees. Using high-resolution data across space and time will allow me to robustly infer a relationship between leaf phenology and growth seasonality.

Outreach: Given the widespread impacts of climate change on ecosystems, understanding how forest communities respond to prolonged growing seasons is crucial. Observing the responses of deciduous tree species to extended seasons may reveal potential benefits for some species and harm for others. These shifts are likely to influence forest dynamics across North America, with potential feedback with future climate change. Therefore, using two experiments and a large-scale remote sensing project, I aim to understand how the growth dynamics of North American tree species will change with longer growing seasons.

References

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I — Contributions to research and development

a. Articles published or accepted in peer-reviewed journals

Wolkovich, E.M., **Rouleau-Desrochers, C.**, Garcia de Cortazar-Atauri, I., Walker, M.A., Lacombe, T. (2025) Building a more predictive model of Terroir for the Anthropocene. *Plants, People, Planet*. DOI: 10.1002/ppp3.70055.
I developed the meta-analysis conceptual figure and led the global mapping analysis that identified key data gaps; I assisted with the writing and edited the manuscript describing methods and implications.

c. Non-peer-reviewed contributions (for example, specialized publications, technical reports, preprints, conference presentations, posters)

Rouleau-Desrochers, C., Patry, É., Gingras, G., Ucler, J., Vézina-Doré, M., Paillé, O., Moghaddam, Y.Y. (2024) "Un été parti en fumée." *Le Point Bio*.
My role as the lead author was to conduct the interviews with researchers and to write the manuscript (work completed as part of my undergraduate studies).

II — Most significant contributions to research and development

1. Predictive model of Terroir under climate change (Plants, People, Planet, 2025). This paper tackles an urgent applied problem: current vine-planting choices and limited genotype×environment data undermine adaptation planning under climate change. I synthesized fragmented genotype×environment datasets into a proposed meta-analysis framework and designed a global data-gap map that reframed the paper's core argument: predictive models must explicitly account for missing data to guide adaptation. Beyond figure preparation, I contributed to shaping the conceptual structure of the manuscript, ensuring that the analyses were presented in a way accessible to both ecologists and practitioners. My work provided the methodological backbone that made the study's recommendations actionable for viticulture planning under climate change.
2. Public communication on Québec wildfires (Le Point Bio, 2024). I wrote and co-produced a public-facing article that explained the causes and health impacts of the 2023 Québec wildfires shortly after the events. Drawing on my boreal-forest ecology background, I led expert interviews and framed complex findings in an accessible jargon. This work reflects my commitment to turning research into actionable public understanding and outreach, and helped me practice rapid, accurate science communication in a high-stakes context.

III — Applicant's statement

Research experience:
The first time I really fell into natural-science research was during the summer of 2021 with Dr. Pierre Drapeau. Over three months, my colleagues and I travelled across Québec investigating woodpecker damage to wooden electrical infrastructures—a project that taught me rigorous field protocols, database management and the basics of scientific coding. In 2022, I held my first Undergraduate Student Research Award (USRA) and carried out intensive fieldwork on Pileated woodpecker habitat in remote boreal forest: long days of banding, nest-searching and tree-classification built my field skillset. Alongside this, I leveraged 20 years of nesting data to understand how Pileated woodpecker nesting phenology is affected by climate change—my first independent statistical project—which gave me hands-on experience in data cleaning, model selection and interpretation of long-term ecological signals. A second USRA at UBC in 2023 (supervised by Drs. Frederik Baumgarten and E.M. Wolkovich) exposed me to large-scale experiments. During the summer, I designed and implemented a defoliation experiment to test tree resilience, maintained

a 1,000-tree trial, deployed dendrometers and analyzed growth responses to treatments. These activities allowed me to frame research questions about climate–growth interactions, refine experimental protocols, and collect data for better predictive models of tree responses under climate change.
In 2024, I conceived and led a greenhouse experiment (foundation of my first MSc chapter), where I formulated hypotheses on plant-climate interactions, designed treatments to test them, and integrated climate and phenological data into statistical models. Alongside this, I also joined field campaigns in Smithers (BC) and Mount Rainier (WA) to study forest regeneration dynamics, where I improved my forestry methods and learned the basics of dendrochronology. In 2025, while continuing my greenhouse experiment—which will later become my first PhD chapter—I visited the Arnold Arboretum of Harvard University to collect tree cores, process cross-sections from a common garden trial, and analyze tree rings. These activities completed the data collection for my second MSc chapter and deepened my expertise in dendrochronological methods. Collectively, these experiences have equipped me with strong skills in experimental design, tree-ring analysis, intensive field logistics, and quantitative methods, which I continue to strengthen through workshops and applied projects.

Relevant activities and distinctions:
Prior to my current academic path, I earned degrees in the restaurant and wine industries and spent nearly ten years working in the field, where I built a strong work ethic and sense of rigour. The transition from the food and beverage industry to academia happened while working for six months in a vineyard in New Zealand; I realized that my ties to nature were stronger than I expected. Thereafter, I enrolled in a Bachelor's degree in biology, majoring in ecology. As an undergraduate, I applied the work ethic I learned in hospitality to my studies and research. In 2025, during the first year of my MSc degree, I deepened my quantitative skills by taking a biomathematics class, attending a three-week workshop on hierarchical Bayesian models and following the Statistical Rethinking course. I also present and refine my models in a biweekly Bayesian working group where peers comment and suggest improvements.
My dedication and constant efforts during my undergraduate studies at Université du Québec à Montréal were recognized with the Governor General's Academic Silver Medal, Dean's List of honours, 13 competitive scholarships totalling $76,750, and a perfect GPA of 4.30/4.30. In addition to my excellent academic record, I have always maintained active outreach: I was interviewed at the 2023 UQÀM scholarship ceremony, featured on Biochambers for my growth-chamber work, presented to Treespotters citizen scientists at Harvard, and hosted an open house at the Canadian Society for Ecology and Evolution (2024).
Across the two years leading my experiment, I developed strong leadership qualities as I trained and supervised 12 undergraduate students in growth-chamber protocols, phenological monitoring and greenhouse management, two of whom are currently pursuing graduate studies. I also led the team for Smithers fieldwork in 2025. In addition to my mentor role in my laboratory, I was a teaching assistant for FRST303 (course enrollment: 1962) and FRST304 (course enrollment: 1216), during which I ran office hours and gave supplementary materials to help students grasp ecological and plant physiology concepts. Outside my lab work, since 2021, I have mentored twelve international students during their first months in Canada, tutored seven peers in biochemistry and statistics, and volunteered with BUDR to advise students on applying to graduate school —often drawing on my own non-linear path. My background in restaurant management was valuable for my greenhouse experiment, as I already had technical and organizational knowledge for successful projects. However, this unique context posed certain challenges related to working with living organisms (e.g. pest management, irrigation control, climate data collection), which I overcame by building a strong network of professors and technicians who could provide me with resources to overcome obstacles. My wildlife photography work complements my outreach: my photos have appeared on the cover of *Le Point Bio (2024)*, in the Biodiversity Research Center (UBC) annual competition, in a recent Proceedings of the National Academy of Sciences (PNAS) article, and in *Le Rappel*.

Natural Sciences and Engineering Research Council of Canada
Conseil de recherches en sciences naturelles et en génie du Canada

Application for a Canada Graduate Research Scholarship - Doctoral or Canada Postdoctoral Research Award (Form 201)

Type of Award		Personal identification no. (PIN)	Family name, given name and initial(s) of applicant
CGRS D		**Valid** 580192	Rouleau-Desrochers, Christophe C

SPECIAL CIRCUMSTANCES

UQAM **Université du Québec à Montréal**
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Transcript

Issued on
July 23, 2024

Rouleau-Desrochers Christophe
6880 RUE MARQUETTES
MONTREAL QC
H2G 2Y8

Permanent code
ROUC22019809

Program **Bachelor's degree in Biology, Problem-Based Learning (6713)**

Term	Code	Credits	Title	Gr. Avg.	Grade
			French Proficiency Examination		S
Courses taken for the Ecology Certificate					
	BIO1050	3	ANIMAL BIOLOGY		A+
	BIO1700	3	BIODIVERSITY CONSERVATION		A+
	BIO2611	3	PLANT BIOLOGY		A+
	BIO3100	3	GENERAL ECOLOGY		A+
	BIO3500	3	ECOLOGY AND DATA ANALYSIS SEMINAR		A+
	BIO6821	3	WILDLIFE MANAGEMENT		A+
F-2021	HIS4592	3	HISTORY OF INDIGENOUS PEOPLE OF CANADA (UP TO THE 19TH CENTURY)		A+
F-2021	PHI3017	2	PHILOSOPHY OF BIOLOGICAL SCIENCES (2 CR.)		A+
Courses taken since admission into the program					
W-2022	BIA1002	1	INTRODUCTION TO THE STUDY OF THE CELL (1 CR.)		S
W-2022	BIA1401	4	DIVERSITY OF MICRO-ORGANISMS (4 CR.)		A+
W-2022	BIA2001	5	CELLULAR BIOLOGY AND BIOCHEMISTRY (5 CR.)		A+
W-2022	BIA2002	4	GENE REPLICATION AND EXPRESSION (4 CR.)		A+
F-2022	BIA2101	5	PLANT METABOLISM AND REGULATION (5 CR.)		A+
F-2022	BIA2201	6	ANIMAL AND HUMAN REGULATION (6 CR.)		A+
F-2022	BIA2501	4	IMMUNOLOGY (4 CR.)		A+
W-2023	BIA1200	2	EVOLUTION AND NATURAL SELECTION (2 CR.)		A+
W-2023	BIA1700	6	ORGANISMS AND THE ENVIRONMENT (6 CR.)		A+
W-2023	BIA2600	4	ENVIRONMENTAL INTERACTION (4 CR.)		A+
W-2023	FSM3300	6	INTRO TO EXPERIMENTAL RESEARCH OR PROFESSIONAL PRACTICE (6 CR.)		A+
F-2023	BIA3010	9	METHODOLOGY IN ECOLOGY (9 CR.)		A+
F-2023	BIA3510	4	SPECIALIZATION I IN ECOLOGY (4 CR.)		A+
F-2023	COM1859	3	ISSUES AND PRACTICES OF SCIENTIFIC COMMUNICATION		A+
W-2024	BIA3511	5	SPECIALIZATION II IN ECOLOGY (5 CR.)		A+

***** END OF TRANSCRIPT *****

Credits awarded: 91.0 GPA: 4.30/4.3

ORDRE DES TRADUCTEURS, TERMINOLOGUES ET INTERPRÈTES AGRÉÉS DU QUÉBEC
Traductrice agréée - Certified Translator
JENNA KHAMIS
33216

Degree: Bachelor of Science (BSc)

Date awarded: June 20, 2024

Certified copy
Registrar
[Signature]
July 23, 2024, by [initials]

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EXPLANATORY NOTES

This note is meant to ensure a better understanding of official student progress as recorded on documents issued in French.

Academic file
Student's academic file, includes admission and registration information and degree conferment data as maintained by the Office of the Registrar, in accordance with existing University policies. Students may request a copy of the information kept in his or her file.

Transcript
A transcript is an official document presenting the cumulative academic record of a student's progress in the programme to which he or she has been admitted. It is prepared by the Office of the Registrar in accordance with the existing University policy.

Certified true copy
Students may request a certified true copy of their transcript or file. A certified true copy is issued by the Office of the Registrar. It bears the signature of the Registrar and the words "copie conforme". This document may be given directly to the student. If a transcript is sent to another institution, it bears the Official University Seal. A transcript with an official seal cannot be sent to the student.

Symbols used in this transcript

Symbol H
"H" indicates that the course does not taken toward degree. Credit is not included in the Cumulative Grade Point Average (CGPA).

Symbol I
"I" indicates that the course is Incomplete. The student has nos yet met the course requirements, and final evaluation cannot be submitted.

Symbol K
"K" indicates the granting of credit based on a prior learning assessment which has been carried out in accordance with existing University regulations. Credit appears on the transcript and academic file.

Symbol L
"L" indicates a failed course or activity that has since been completed.

Symbol P
"P" indicates a prerequisite course. Grade is not included in CGPA.

Symbol R
"R" indicates that the grade is to be assigned at a later date.

Symbol W
"W" indicates an audit course or activity. Grade is not included in CGPA.

Symbol X
"X" indicates withdrawal with permission. Credit does not appear on the transcript or academic file.

Grade of S
"S" indicates that the course requirements have successfully completed. Credit granted, but not included in CGPA.

Grade of V
"V" indicates that the course has successfully completed at another institution under a specific agreement. Credit granted, but not included in CGPA.

Credit
Numerical value on the work that student must complete to attain specific course objectives.

Credits earned
"Credits earned" (crédits réussis) is the total number of credits earned towards fulfilment of programme requirements. It includes credits for successfully completed courses and those granted by advanced standing (equivalence). It excludes credit for prerequisite course work (P) and for courses not taken towards the degree (H).

Grading System prior to Fall 1992
A, B, C, and D.

Grading System as of Fall of 1992
A+, A, A-, B+, B, B-, C+, C, C-, D+, D.

Failed Course
A student who does not meet course requirements receives the letter "E" as the grade for that course. Credits for that course do not appear on the transcript.

Credit from another programme within the University
The words "Activité suivie dans le cadre . . . " indicate that the activity has been completed while completing requirements for another programme. Credit for this activity has been assigned to the student's current programme.

Transfer Credits
The words "X Crédits intégrés au programme" indicate that the student has met certain programme requirements prior to admission.

Legend
ABS : Absence EXE : Advanced standing N.A. : Not applicable

Cumulative Grade Point Average

The cumulative grade point average (CGPA) is calculated as follows :

$$M = \frac{\sum PiCi}{\sum Ci}$$

Signification des symboles :

$\sum$ = Combined total of
P = Numeric value assigned to letter grades
C = Number of credits assigned to courses for which the letter grade has a numeric value
i = Specific course or activity
M = Average

Numeric Values of Grades :

A+ = 4,3	A = 4,0	A- = 3,7 ;	Excellent
B+ = 3,3	B = 3,0	B- = 2,7 ;	Very good
C+ = 2,3	C = 2,0	C- = 1,7 ;	Good
D+ = 1,3	D = 1,0 ;		Pass
E = 0 ;			Fail

Calculating the CGPA enroled prior to September 1992

A new grading system was introduced in September 1992 (see grading system above). In accordance with policies accompanying the change, the CGPA of students who began their programme before September 1992 is calculated as follows :

$$M = \frac{\left\{ \left(\sum PiCi \textit{ calculated before September 1992} \right) \times \frac{4,3}{4,0} \right\} + \left\{ \sum PiCi \textit{ calculated after September 1992} \right\}}{\sum Ci}$$

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H3C 3P8

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Translator's Declaration

Document: Université du Québec à Montréal Official Transcript—Christophe Rouleau-Desrochers

I, the undersigned, Jenna Khamis, Certified Translator from French to English and member in good standing of the *Ordre de traducteurs, terminologues et interprètes agréés du Québec* (OTTIAQ) (member no. 33216), hereby certify hereby declare that I did translate into English the attached French document and that, to the best of my knowledge, the translation accurately reflects the contents and meaning of the French original.

The translation was completed in accordance with the *Règles de pratique professionnelle de l'Ordre des traducteurs, terminologues et interprètes agréés du Québec (OTTIAQ)*. Under these rules, the certified translator is required to comply with OTTIAQ's code of ethics and to follow a rigorous process when completing and revising the translation to provide reasonable assurance that the translation is an appropriate equivalent to the source text.

This certification is valid provided that no changes are made to the translation without my prior approval.

Certified at Montreal, Quebec on November 6, 2024

Jenna Khamis, C. Tr.

ORDRE DES TRADUCTEURS, TERMINOLOGUES ET INTERPRÈTES AGRÉÉS DU QUÉBEC
Traductrice agréée - Certified Translator
JENNA KHAMIS
33216

UQÀM **Université du Québec à Montréal**
Registrariat

RELEVÉ DE NOTES

Date d'émission
23 juillet 2024
Code permanent
ROUC22019809

Rouleau-Desrochers Christophe
6880 RUE MARQUETTE
MONTREAL QC
H2G 2Y8

Programme : **baccalauréat en biologie en apprentissage par problèmes (6713)**

Trimestre	Sigle	Crédits	Titre	Moyenne du groupe	Résultat
			Test de maîtrise du français		S
Activités suivies dans le cadre du certificat en écologie					
	BIO1050	3	BIOLOGIE ANIMALE		A+
	BIO1700	3	CONSERVATION DE LA BIODIVERSITE		A+
	BIO2611	3	BIOLOGIE VEGETALE		A+
	BIO3100	3	ECOLOGIE GENERALE		A+
	BIO3500	3	SEMINAIRE EN ECOLOGIE ET ANALYSE DES DONNEES		A+
	BIO6821	3	AMENAGEMENT DE LA FAUNE		A+
AUT.2021	HIS4592	3	HISTOIRE DES AUTOCHTONES DU CANADA (JUSQU'AU XIX S.)		A+
AUT.2021	PHI3017	2	PHILOSOPHIE DES SCIENCES BIOLOGIQUES (2 CR.)		A+
Activités suivies depuis l'admission au programme					
HIV.2022	BIA1002	1	INITIATION A L'ETUDE DE LA CELLULE (1 CR.)		S
HIV.2022	BIA1401	4	DIVERSITE DES MICROORGANISMES (4 CR.)		A+
HIV.2022	BIA2001	5	BIOCHIMIE ET BIOLOGIE CELLULAIRE (5 CR.)		A+
HIV.2022	BIA2002	4	REPLICATION ET EXPRESSION DES GENES (4 CR.)		A+
AUT.2022	BIA2101	5	METABOLISME ET REGULATION DES VEGETAUX (5 CREDITS)		A+
AUT.2022	BIA2201	6	REGULATION ANIMALE ET HUMAINE (6 CR.)		A+
AUT.2022	BIA2501	4	IMMUNOLOGIE (4 CR.)		A+
HIV.2023	BIA1200	2	EVOLUTION ET SELECTION NATURELLE (2 CR)		A+
HIV.2023	BIA1700	6	ORGANISMES ET ENVIRONNEMENT (6 CREDITS)		A+
HIV.2023	BIA2600	4	INTERACTION ENVIRONNEMENTALE (4 CR)		A+
HIV.2023	FSM3300	6	INIT.A LA RECHERCHE EXPERIM.OU PRAT.PROF.(6CR)		A+
AUT.2023	BIA3010	9	METHODOLOGIE EN ECOLOGIE (9 CR)		A+
AUT.2023	BIA3510	4	SPECIALISATION I EN ECOLOGIE (4 CR)		A+
AUT.2023	COM1859	3	ENJEUX ET PRATIQUES DE LA COMMUNICATION SCIENTIFIQUE		A+
HIV.2024	BIA3511	5	SPECIALISATION II EN ECOLOGIE (5 CR.)		A+

---------- FIN DU RELEVÉ ----------

Crédits réussis : 91.0

Moyenne cumulative : 4.30/4.3

Grade : bachelier ès sciences (B.Sc.)

Date d'obtention : 20 juin 2024

Copie conforme au dossier
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23 juillet 2024, par

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THE UNIVERSITY OF BRITISH COLUMBIA

ADVISING TRANSCRIPT

ENROLMENT SERVICES
2016- 1874 East Mall
Vancouver, BC Canada V6T 1Z1

Surname:	Given Name:	Student Number:	Date:
Rouleau-Desrochers	Christophe	49411424	August 29, 2025

Note: Advising transcript is an unofficial record of your UBC academic history.

UBC Credentials

None to date

Transfer Credits

None to date

2024-25 Winter Session

M.Sc., In Forestry (Vancouver)

Term	Course		Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	BIOL_V	301	(3.0)	Biomathematics	78	B+	3.0	46	82
1-2	FRST_V	549B	(12.0)	Master's Thesis		T			
2	CONS_V	310	(3.0)	Ecology in a Changing Climate	89	A	3.0	30	81
2	FRST_V	507C	(3.0)	Topics in Forest Science	87	A	3.0	11	91

Cumulative Totals
Credits Attempted: 21.0 Credits Earned: 9.0 Average: 84.7%

Sessional Average: 84.7%

Credits Attempted	=	Passed	Failed	Withdrawn	Incomplete
21.0		9.0	0.0	0.0	12.0

2025 Summer Session

M.Sc., In Forestry (Vancouver)

Term	Course		Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
	FRST_V	549C	(18.0)	Master's Thesis					

Cumulative Totals
Credits Attempted: 39.0 Credits Earned: 9.0 Average: 84.7%

Sessional Average:

Credits Attempted	=	Passed	Failed	Withdrawn	Incomplete
18.0		0.0	0.0	0.0	18.0

UBC Academic Awards
NSERC Canada Graduate Scholarship Master's Award

2025-26 Winter Session

M.Sc., In Forestry (Vancouver)

Term	Course		Credit Value	Course Title	% Grade	Grade	Credit Rec'd	Class Size	Class Avg
1	FRST_V	505C	(3.0)	Directed Studies in Forest Science					
1-2	FRST_V	549C	(18.0)	Master's Thesis					

Cumulative Totals
Credits Attempted: 60.0 Credits Earned: 9.0 Average: 84.7%

Sessional Average:

Credits Attempted	=	Passed	Failed	Withdrawn	Incomplete
21.0		0.0	0.0	0.0	21.0

UBC Academic Awards
NSERC Canada Graduate Scholarship Master's Award

********** End of Record **********

THE UNIVERSITY OF BRITISH COLUMBIA

The University of British Columbia was established by an Act of the Provincial Legislature in 1908 and opened in 1915. The University is a member of the Association of Universities and Colleges of Canada. UBC Okanagan was established as a campus of The University of British Columbia in April 2005. UBC Okanagan assumed responsibility for several Okanagan University College degree programs and all continuing students in those programs on July 1, 2005. Okanagan University College ceased its educational operations as of August 31, 2005.

EXPLANATION OF STUDENT ACADEMIC RECORD

Academic terms

The University operates on a year system. The academic year is divided into two sessions. Winter Session is divided into two terms, Term 1 from September to December and Term 2 from January to April. Summer Session starts in May and ends in August and is comprised of Term 1 (May and June) and Term 2 (July and August). Courses may span a single term (Term 1 or 2) or a full session (Term 1 and 2) for either session.

Distance Education and Technology (DET) courses begin on one of two starting dates during Winter Session, designated in the Term column as A,B,C or D. During the Summer Session, DET courses have only one starting date and are designated with an A or B in the Term column.

Academic Awards

All academic awards (Fellowships, Scholarships, Prizes and Medals) administered by the University and issued subsequent to April 1975, are noted on the transcript of academic record.

Course Numbering

Courses numbered 100 to 199 are primarily for first-year students. Similarly, courses numbered 200 to 299 are primarily for second-year students, courses numbered 300 to 399 are for third-year students, and courses numbered 400 to 499 are for fourth-year students. Courses numbered 500 to 699 are considered graduate-level. Courses numbered 700 and above are clinical courses.

Class Size and Class Average Grades

Effective February 1999, class size and class average grades are shown for most courses. In courses with more than one section, the class size and average are reported for the student's section.

Grading and Course Weight

In May of 1991, The University of British Columbia adopted a percentage grading and credit weighting system. Course weight is expressed in "credits". In general 1 credit represents 1 hour of instruction or 2 to 3 hours of laboratory work per week throughout one term of the Winter Session. A 1-unit course became a 2-credit course. Courses are normally graded on a percentage basis with a corresponding letter grade derived as shown.

Degree Standings and Averages

Standings are expressed as Class 1 (80% or over), Class 2 (65 to 79%), Class P (50 to 64%) or Honours; not used by all faculties. Effective Winter Session 1996/97, the calculation of the standing and average is based on the average percentage grade of all upper-level (300 or higher) courses used to satisfy the degree requirements (excluding courses graded as Pass/Fail); not used by all faculties. This is not applicable to programs in the Faculty of Graduate Studies.

Undergraduate Grading Scale

Percent	Letter Grade	Percent	Letter Grade
90-100	A+	64-67*	C+
85-89	A	60-63*	C
80-84	A-	55-59*	C-
76-79	B+	50-54*	D
72-75	B	0-49	F* (Fail)
68-71	B-		

Graduate Grading Scale

Masters Grading Scale		Doctoral Grading Scale	
Percent	Letter Grade	Percent	Letter Grade
90-100	A+	90-100	A+
85-89	A	85-89	A
80-84	A-	80-84	A-
76-79	B+	76-79	B+
72-75	B	72-75	B
68-71	B-	68-71	B-
64-67	C+	0-67	F* (Fail)
60-63	C		
0-59	F* (Fail)		

*Some programs may require higher passing grades; failing grades are assigned a letter grade of "F."

The following standings also apply:

- AEG - aegrotat standing: indicates that the student is granted credit although unable to complete because of illness. A letter grade will also be recorded and that grade converted to a minimum percentage grade for that category for the calculation of averages.
- AUD - audit
- CH - challenge credit
- CIP - course in progress
- CR/D/F - (introduced in 2010/11) excluded from calculation in all averages; granted where a grade is assigned by an instructor but not included on the student's official transcript; grade is converted to CR, D or F standing. CR- credit, satisfactory completion of the requirements of the subject; D- marginal pass (50%- 54%), and not available for those programs or courses that define a passing grade as higher than 50%; F - fail
- EX - exempt
- J - see "A" standing as defined in grading scale prior to May 1991
- P or F - no grade assigned, excluded from calculation in all averages; P- pass, requirements of subject completed satisfactorily, credit granted where applicable; F - fail
- H - (introduced 1999/00 session) no grade assigned, excluded from calculation in all averages; H - honours, exceeds course requirements, credit granted where applicable
- PLA - credit assigned based on prior learning in subject area.
- S - see "S" standing as defined in grading scale prior to May 1991
- SD - standing deferred; excluded from calculation in all averages
- T - thesis in progress; graduating essay not submitted; course continuing
- W - official withdrawal

PRIOR TO MAY 1991

Course Weight and Hours

Course weight was expressed in "units". In general, 1 unit represented 1 lecture hour per week (or a laboratory period of approximately 2 to 3 hours each week) throughout both terms of the Winter Session. Some faculties did not assign a unit value to some courses. The maximum mark possible (max) indicated the weight of a course. A 3-unit course (3 hours of lectures per week throughout the Winter Session) was graded out of a total of 150 marks. This course approximated a 6 semester-hour or a 9 quarter-hour course.

Grading classifications

- Class 1 - 80% or better (i.e., 120 or more on a maximum of 150)
- Class 2 - 65%-79% (i.e., 98 to 119 out of 150)
- Class P - 50%-64% (i.e., 75 to 97 out of 150) except Dentistry, Graduate Studies, Librarianship, Medicine, Nursing and Rehabilitation Medicine where the range is 60%- 64%
- Class F - failure, no credit granted: below 50% except Dentistry, Graduate Studies, Librarianship, Medicine, Nursing and Rehabilitation Medicine: below 60%

The following standings also apply:

- A - adjudicated pass: indicates that credit is granted and the course need not be repeated although it may not normally qualify as a prerequisite for further work. The grade assigned by the instructor is used in calculation of averages.
- AEG - aegrotat standing: granted to a student who has been successful in studies but unable to write the final exam because of illness or other extenuating circumstances
- AUD - audit
- C - requirements of subject completed satisfactorily; no grade assigned; unit value granted where applicable; excluded from calculation in all averages
- D - standing deferred; excluded from calculation in all averages
- E - exempt
- N - no credit granted; did not write examination or otherwise complete requirements of the course (discontinued beginning 1985- 86 session)
- S - supplemental privilege granted; failure; no credit granted; supplementals are not permitted in all faculties
- T - thesis in progress; graduating essay not submitted; course continuing
- W - (introduced in 1988/89 session) official withdrawal
- X - time expired; student did not complete course

Year Standing

In most Faculties each "year" was graded as Class 1, Class 2, Class P, Fail or carried a descriptive clause indicating action taken by the Adjudication Committee.

Degree Standing

Expressed as Class 1, Class 2, Class P, or with Honours; not used by all faculties. In some faculties the calculation of standing was based on the studies of the final two years, in others on the final year only.

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